FEDERAL STANDARD

TAPE, VIDEO, MAGNETIC, RECORDING, FORMATS FOR

Authority: This standard is issued pursuant to the Federal Property and Administrative Services Act of 1949, as amended, and its application to both magnetic video tape recording and magnetic video tape recording/reproducing equipment is mandatory on all Federal agencies.

S1. Purpose and scope. This standard provides detail recording formats for magnetic video tape recordings that are necessary to provide for interchangeability among tapes recorded by using the same format. Adherence to this standard will also minimize the number and types of formats that are to be used.

S1.1 Classification. This standard covers video tape formats and their applications by all Federal agencies in the recording of magnetic video tape.

S1.2 Format identification. The standard video recording formats shall be identified by type, as follows:

Type I - Two-inch Quadruplex, Reel-to-Reel, Format for.
Type II - One-inch Helical Scan, Reel-to-Reel, Format for.
Type III - Three-quarter-inch Helical Scan, Cassette, Format for.
Type IV - One-half-inch Helical Scan, Reel-to-Reel, Format for.

S2. Application. This standard shall be used by all Federal Agencies which use such equipment for entertainment, education and training. This standard does not apply to formats used for special applications, tactical use, and instrumentation recordings. Procurement of such magnetic video tape recording and reproducing equipment shall be limited to equipment that records and reproduces in accordance with the recording formats and other requirements given herein. Existing Government stocks of equipment using other recording formats may continue to be utilized until exhausted.

S3. Referenced documents. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bid or request for proposal shall apply.

American National Standards Institute (ANSI) Publications:

C98.1 - 1963 (R 1969) - Dimensions of 2-inch Video Magnetic Tape.
C98.4 - 1970 - Speed of 2-inch Tape for Quadruplex Video Magnetic Tape Recording.
C98.6 - 1973 - Video, Audio and Tracking Control Records on 2-inch Video Magnetic Tape Quadruplex Recorded at 15 and 7.5 inches per second, Dimensions of.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

Society of Motion Picture and Television Engineers (SMPTE) Recommended Practices:

RP-16 - September, 1970 - Specifications of Tracking Control Record for 2-inch Quadruplex Video Magnetic Tape Recordings.

(Application for copies should be addressed to the Society of Motion Picture and Television Engineers, 862 Scarsdale Avenue, Scarsdale, NY 10583.)

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Electronic Industries Association of Japan (EIAJ):

CP-504 - Interchangeability of Recorded Magnetic Video Tapes for Video Tape Recorders, Magnetic Tape Format.
CP-504 - Supplement - Interchangeability of Recorded Magnetic Video Tapes for Video Tape Recorders - FM Carrier Frequency and Emphasis Characteristics.
CP-507 - Interchangeability of Recorded Magnetic Video Tapes for Video Tape Recorders - Magnetic Video Recording for Color Television Signal.

(Application for copies should be addressed to Electronic Industries Association of Japan (EIA-J), ATTN: Mr. William Tanaka, Federal Bar Building West, 1819 H. Street, NW, Washington, DC 20006.)

Electronic Industries Association (EIA) Standards:

Standard RS-170 - November, 1957 - Electrical Performance Standards - Monochrome Television Studio Facilities.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 11 West 42nd Street, New York, NY 11232.)

S4. General characteristics.

S4.1 Scanning. Except as otherwise specified in the individual recording format, the standard American television nominal scanning rate of 30 frames per second, with two 262.5 horizontal-line fields interlaced in each frame, shall apply to the recording formats specified herein. Tolerance in dimensions will be found on referenced figures and tables.

S4.2 Monochrome and color recording. The recording formats specified herein are applicable for both monochrome and color video recording.

S4.3 Type I format shall conform to the ANSI and SMPTE standards referenced in section 3.

S4.4 Type II format shall conform to figure 1 of this document.

S4.5 Type III format shall conform to figure 2 and table I of this document. The cassette shall conform to figures 3, 4, 5, 6, and 7 of this document.

S4.6 Type IV format shall conform to figure 8 and table II of this document, and to EIAJ standards referenced in section 3.

S5. Detailed format characteristics.

S5.1 Type I format. The type I recording format shall be in accordance with the following:

S5.1.1 Tape speed. Tape speed shall be 38.1 centimeters per second (15 inches per second) in accordance with ANSI C98.4.

S5.1.2 Tape dimensions. Tape dimensions shall be in accordance with ANSI C98.1.

S5.1.3 Video, audio and tracking control records locations and dimensions. The video, audio, and tracking control records locations and dimensions shall be in accordance with ANSI C98.6.

S5.1.4 Tracking control record. The relationships among the tracking control record, frame pulse record, and video record shall be in accordance with SMPTE Recommended Practice RP-16.

S5.2 Type II - Recording format for one-inch, helical scan video tape. The type II recording format shall be in accordance with the requirements specified by the procuring activity.

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S5.3 Type III - Recording format for three-fourths-inch, helical scan, U-cassette, video tape. The type III recording format and the cassette dimensions shall conform to the following requirements:

S5.3.1 Format standards.

(a) Video scanning system: 2-head slant-track recording.
(b) Tape width: 19.00 millimeters ± 0.03 millimeter.
(c) Tape speed: 95.3 millimeters per second ± 0.2 millimeter per second.

S5.3.2 Tape format. The magnetic format (pattern of tracking on the tape) is to be as shown in figure 2 and table 1. Figure 2 shows the format as viewed from the magneto-sensitive side of the tape. General track requirements are as follows:

(a) Recordings shall be arranged from top edge to bottom edge as follows: control track, video tracks, audio track 2, audio track 1.
(b) Audio shall be recorded on two independent tracks for stereo or bilingual applications.
(c) Video tracks shall be arranged so that one full field of video information is recorded on each track.
(d) Control track shall be recorded in the form of a 29.97 Hz signal.

S5.3.2.1 Video track angle. The angle between direction of tape motion and the plane in which the video head rotates shall be 4 degrees 54 minutes 49.1 seconds. (NOTE: This angle is for stationary tape.)

S5.3.2.2 Video head switching. Switching between video heads during playback shall occur 5 to 8 horizontal lines before vertical sync.

S5.3.2.3 Audio control-head position. The position of the audio/control track head shall be immediately before the capstan drive after tape comes off the supply reel.

S5.4 Type IV - Recording format for one-half-inch, helical scan, video tape. The type IV recording format shall conform to the following requirements.

S5.4.1 Tape speed. Tape speed shall be 190.5 millimeters (7.5 inches) per second in accordance with EIAJ Standard CP-504 and in accordance with figure 8 and table II of this document.

S5.4.2 Tape dimensions. The video tape for this format shall be 12.7 millimeters (0.5 inch) wide and a full reel of tape shall be of sufficient length to record or play back video information for at least 63 minutes at a tape speed of 190.5 millimeters (7.5 inches) per second.

S5.4.3 Video, audio and tracking control records, locations, and dimensions. The video, audio, and tracking control records locations and dimensions shall be in accordance with EIAJ Standard CP-504 and figure 8.

S5.4.4 Control track signal, frequency, phase and waveform. The tracking control signal shall be directly recorded on the control track at a higher level than that required for saturation. The waveform and signal polarity of the control track-signal shall be as shown in EIAJ Standard CP-504. The frequency shall be a repeating waveform of 29.97 Hz.

S5.4.5 Color requirements. The color requirements for magnetic video recording shall be in accordance with EIAJ Standard CP-507.

S6. Changes. When a Federal agency considers that this standard does not provide for its essential needs, written request for changing or adding to the standard, supported by adequate justification explaining wherein the standard does not provide for essential needs, shall be sent in duplicate to the General Services Administration, Federal Supply Service, Office of Standards and Quality Control, Washington, DC 20406. The General Services Administration will determine the appropriate action to be taken and will notify the agency.

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S7. Conflict with referenced documents. Where the requirements stated in this standard conflict with any requirement in a referenced document, the requirements of this standard shall apply. Nature of conflict between the standard and the referenced document shall be submitted in duplicate to the General Services Administration, Federal Supply Service, Office of Standards and Quality Control, Washington, DC 20406.

S8. Supersession data. This standard supersedes Military Standard MIL-STD-1856A, 17 June 1975.

Military Custodians:

Army - EL
Navy - EC
Air Force - 80

Review Activities:

Army - EL
Navy - EC
Air Force - 80

User Activities:

Army
Navy
Air Force

Preparing Activity:

GSA-FSS

Civil Agency Coordinating Activities:

Commerce
CSC
DOT
EPA
Energy Research Dev. Adm.
FCC
GSA-FSS
HEW
HUD
Interior
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USIA
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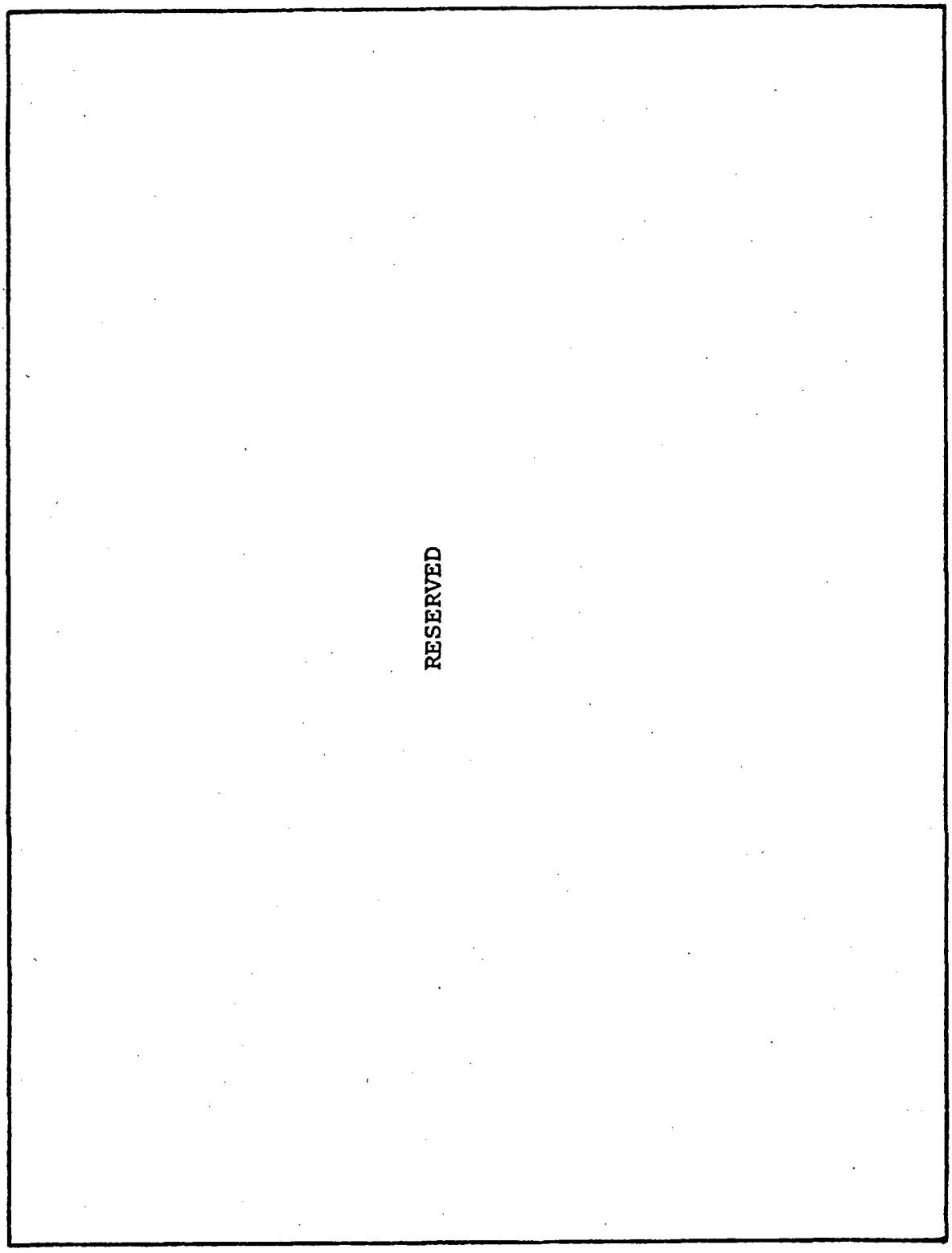


Figure 1. ONE-INCH, HELICAL SCAN, REEL-TO-REEL, FORMAT FOR. (RESERVED)

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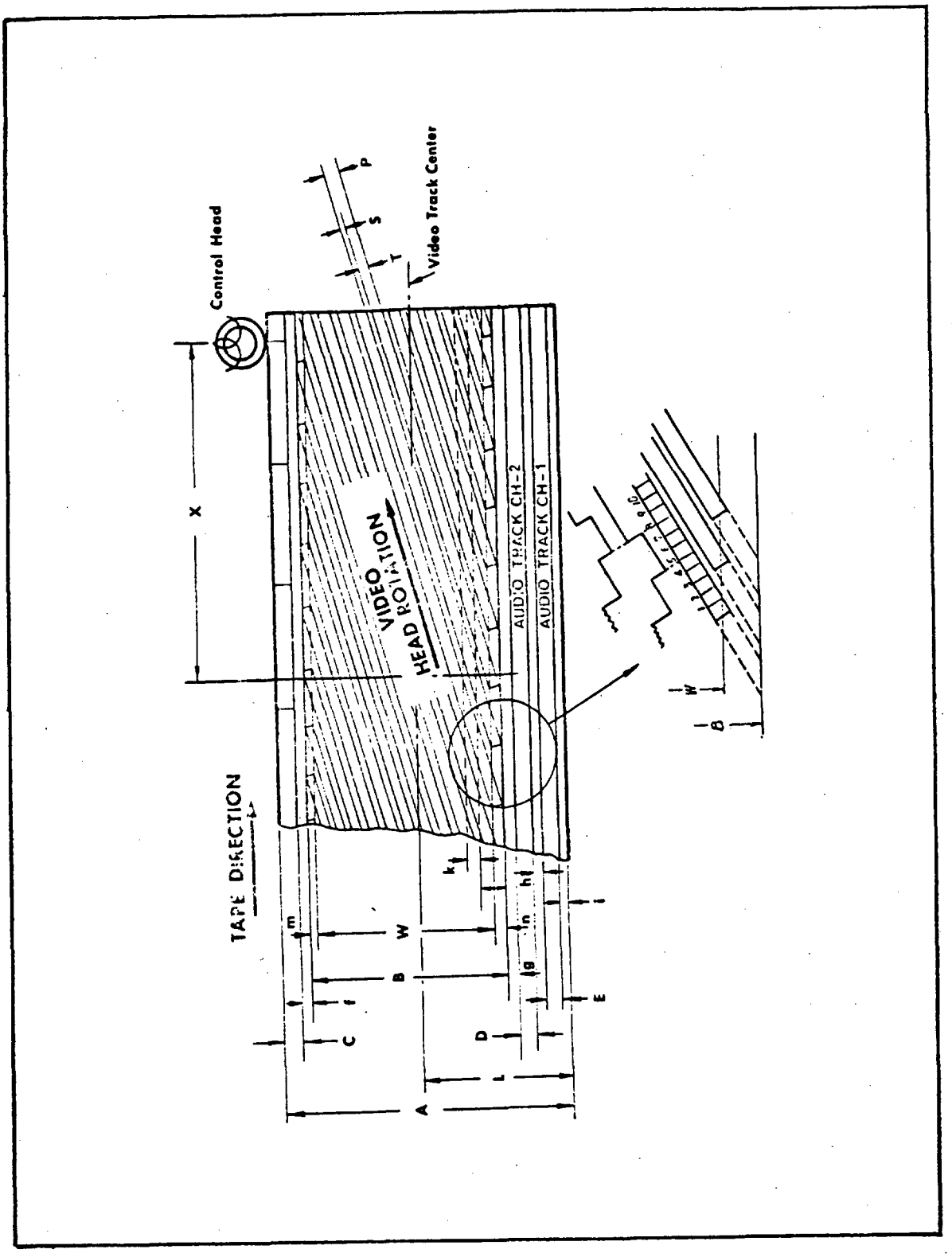


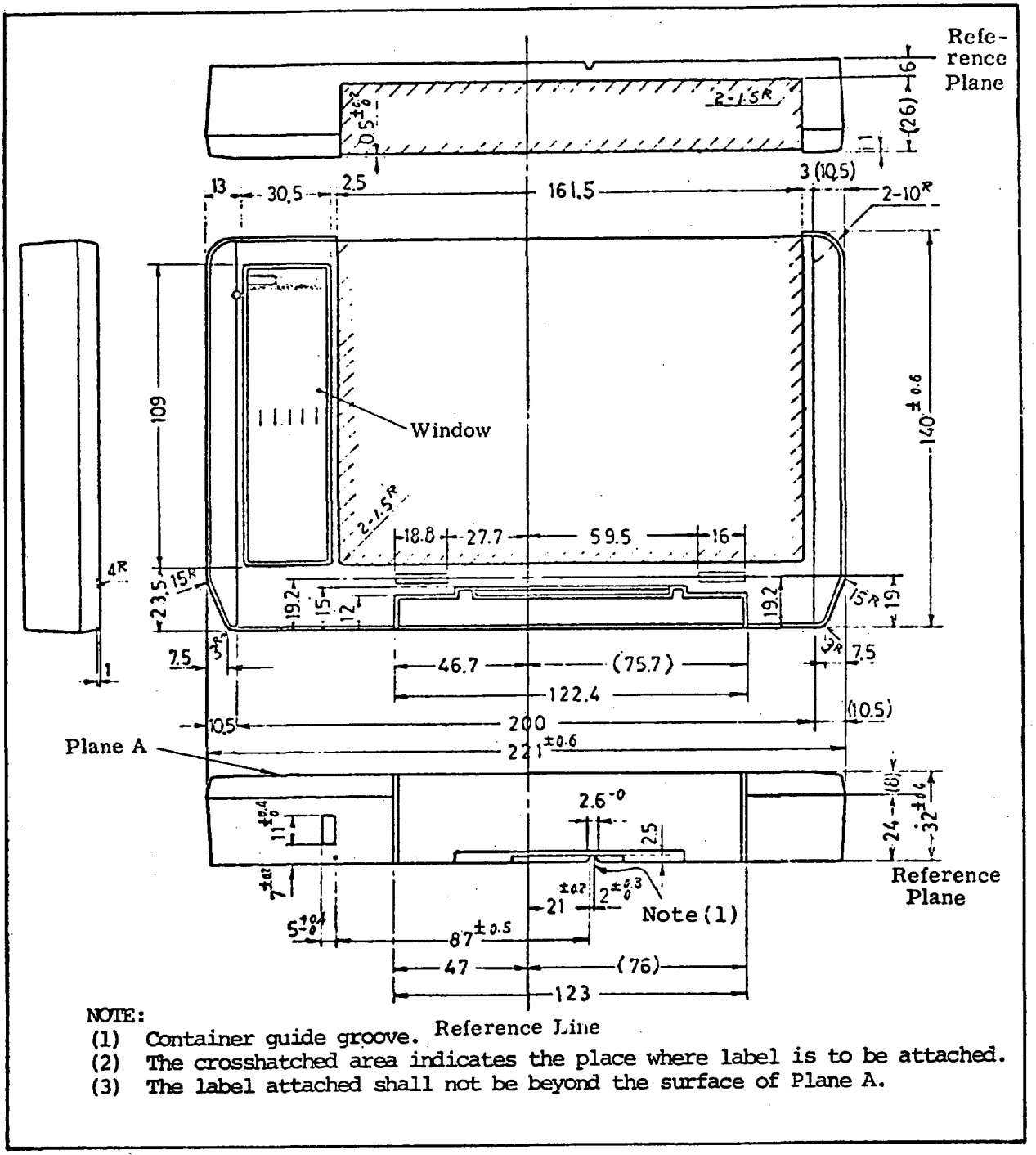
Figure 2. THREE-FOURTHS-INCH HELICAL SCAN, CASSETTE, FORMAT FOR.

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TABLE I

Symbol	Item	
A	TAPE WIDTH	19.00 mm ± 0.03
Vt	TAPE SPEED	95.3 mm/s ± 0.2%
Vr	VIDEO WRITING SPEED	10.26 m/s
P	VIDEO TRACK PITCH	0.1374 mm
B	VIDEO RECORDING ZONE	15.5 mm
W	VIDEO RECORDING ZONE EFFECTIVE WIDTH	14.8 mm
L	CENTER OF VIDEO TRACK	10.450 ± 0.05
T	VIDEO TRACK WIDTH	0.085 ± 0.007
C	CONTROL TRACK WIDTH	0.6 mm
D	AUDIO CH-2 TRACK WIDTH	0.8 ± 0.05 mm
E	AUDIO CH-1 TRACK WIDTH	0.8 ± 0.05 mm
S	VIDEO - VIDEO GUARD BAND	0.052 mm
f	CONTROL - VIDEO GUARD BAND	0.2 mm
g	AUDIO - VIDEO GUARD BAND	0.2 mm
h	AUDIO - AUDIO GUARD BAND	0.7 mm
i	EDGE - AUDIO GUARD BAND	0.2 ± 0.1mm
m	BACK OVERLAP	0.35 mm
n	FRONT OVERLAP	0.35 mm
o	VIDEO TRACK ANGLE	
	TAPE STATIONARY	4° 54' 49.1"
	TAPE MOVING	4° 57' 33.2"
	VIDEO TRACK LENGTH	171.2 mm
k	CUE TRACK WIDTH	0.5 ± 0.05 mm
q	CUE TRACK TO LOWER LIMIT OF VIDEO TRACKS	1.1 ± 0.05 mm
X	DISTANCE FROM THE END OF 180° SCAN OF VIDEO HEAD TO AUDIO AND CONTROL HEAD POSITION ON THE TAPE	74 mm

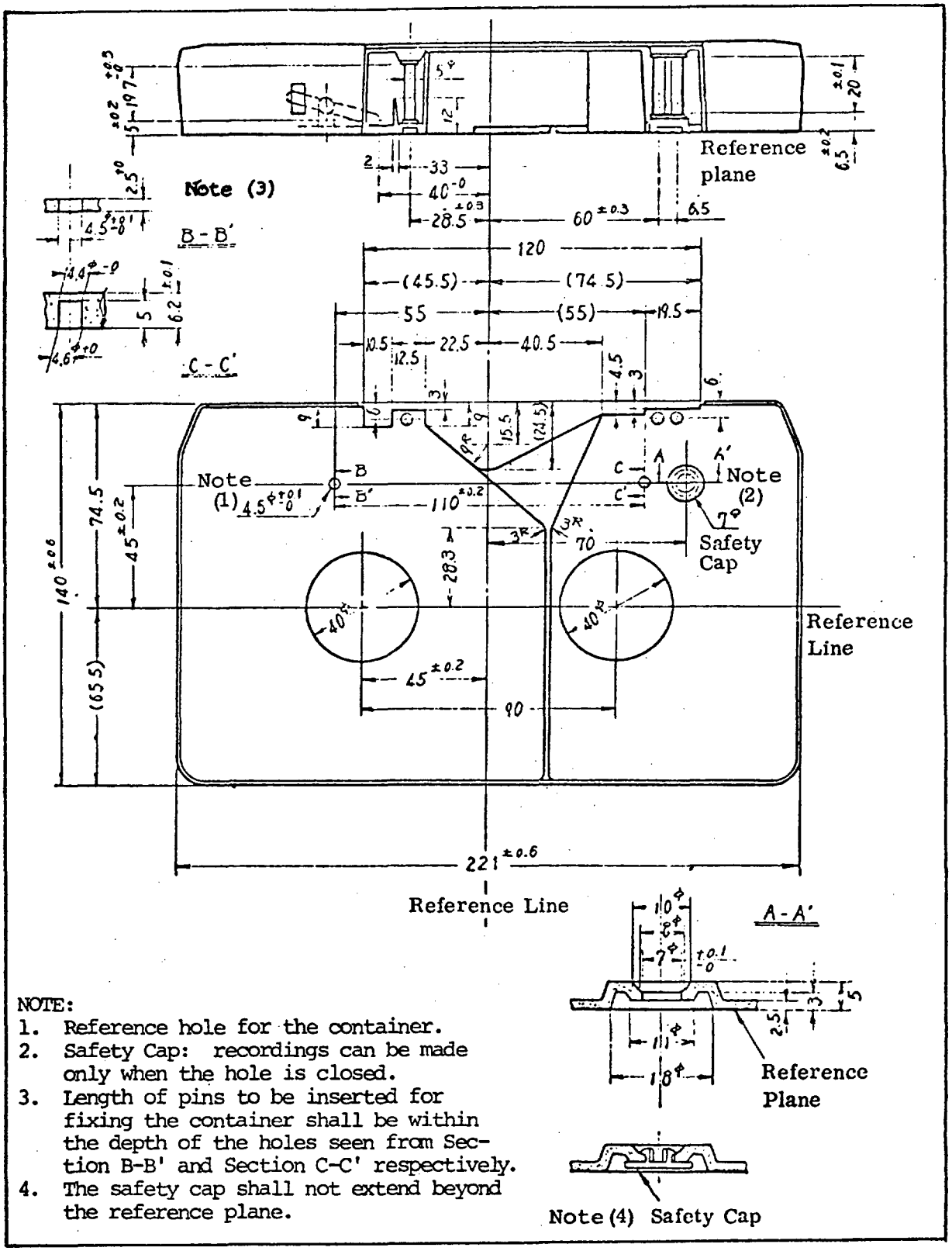
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NOTE:
(1) Container guide groove. The crosshatched area indicates the place where label is to be attached.
(2) The label attached shall not be beyond the surface of Plane A.

Figure 3. THREE-FOURTHS-INCH MAGNETIC TAPE CASSETTE.

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NOTE:

1. Reference hole for the container.
2. Safety Cap: recordings can be made only when the hole is closed.
3. Length of pins to be inserted for fixing the container shall be within the depth of the holes seen from Section B-B' and Section C-C' respectively.
4. The safety cap shall not extend beyond the reference plane.

Figure 4. THREE-FOURTHS-INCH MAGNETIC TAPE CASSETTE.

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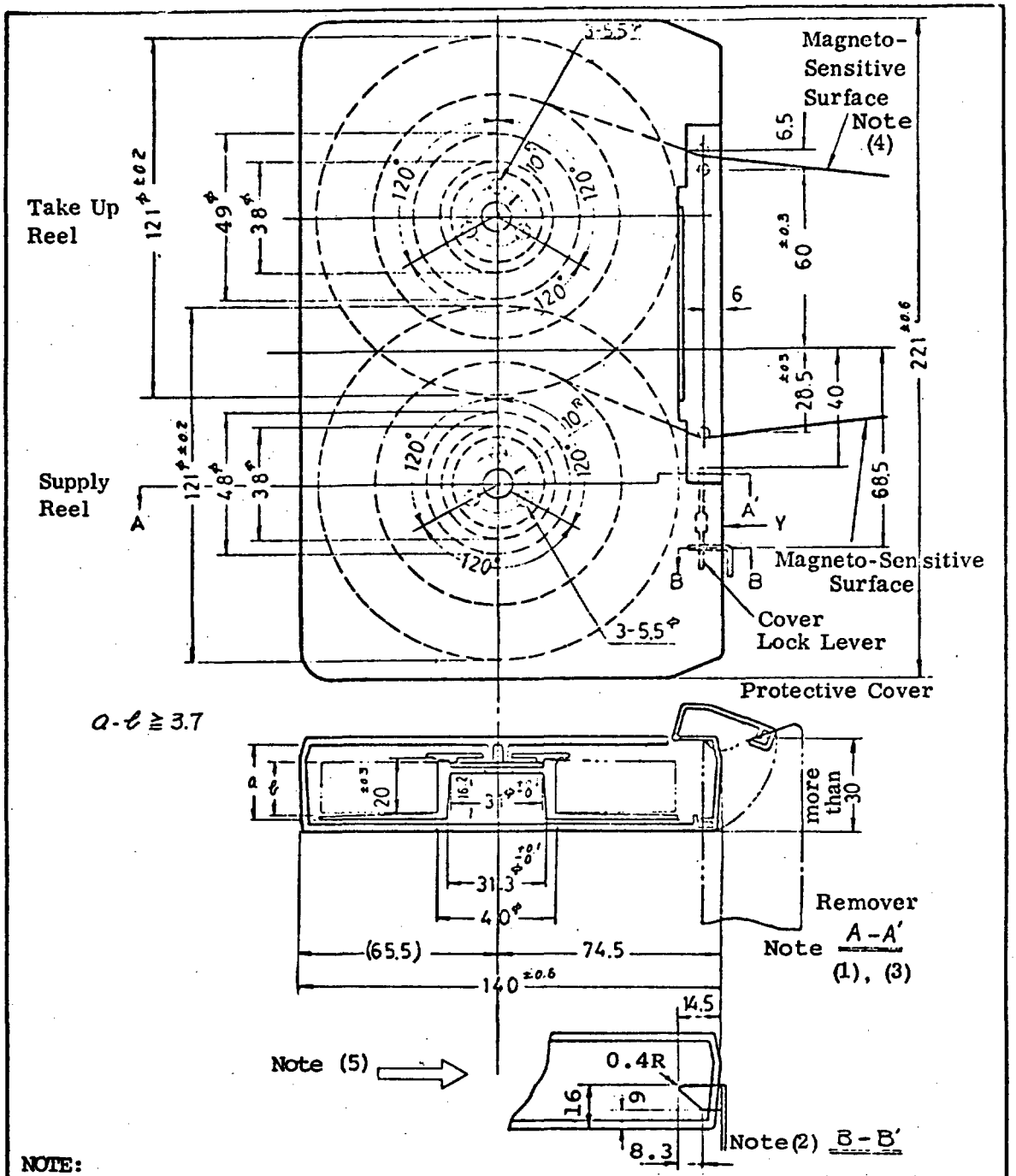


Figure 5. THREE-FOURTHS-INCH MAGNETIC TAPE CASSETTE.

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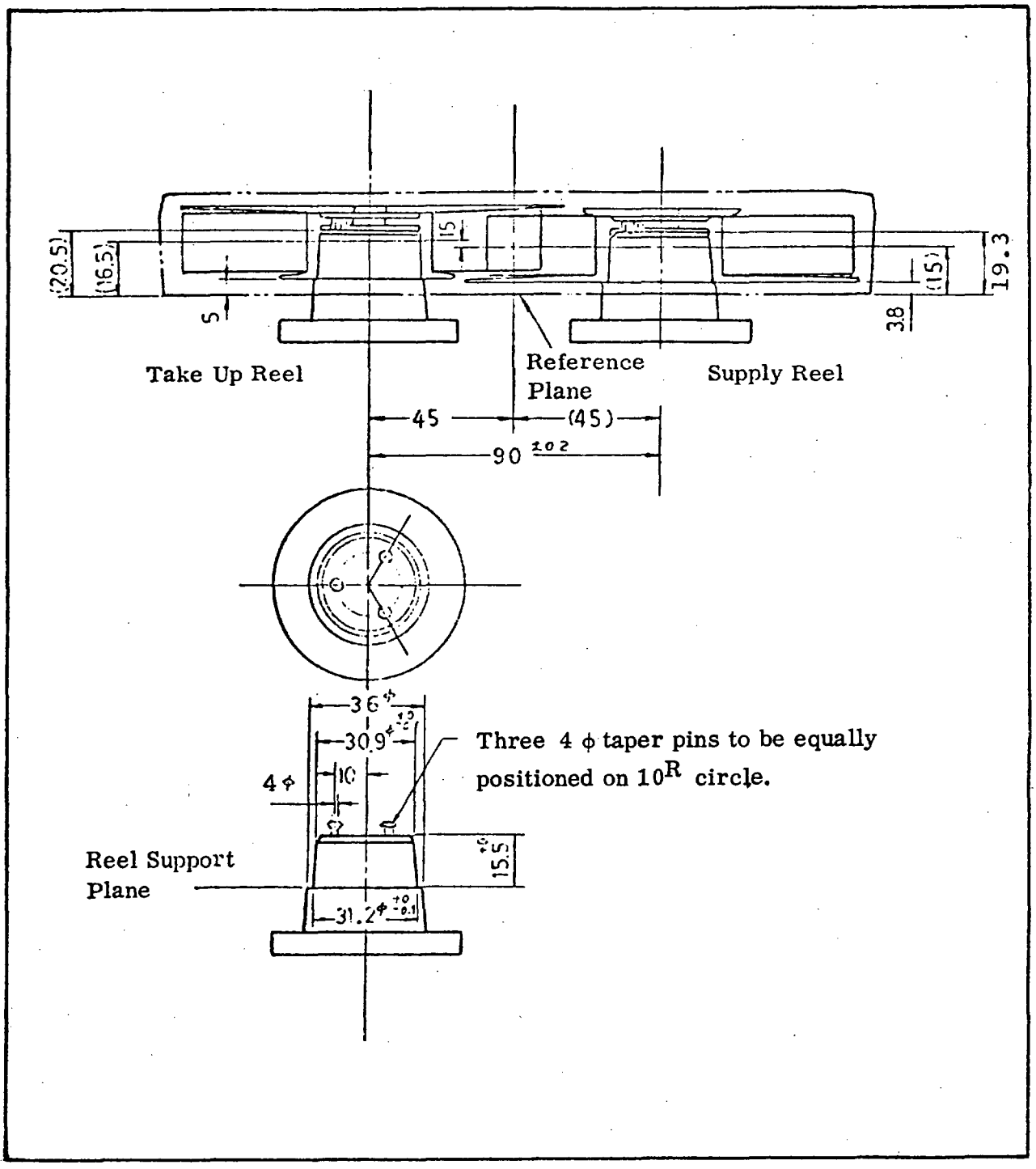
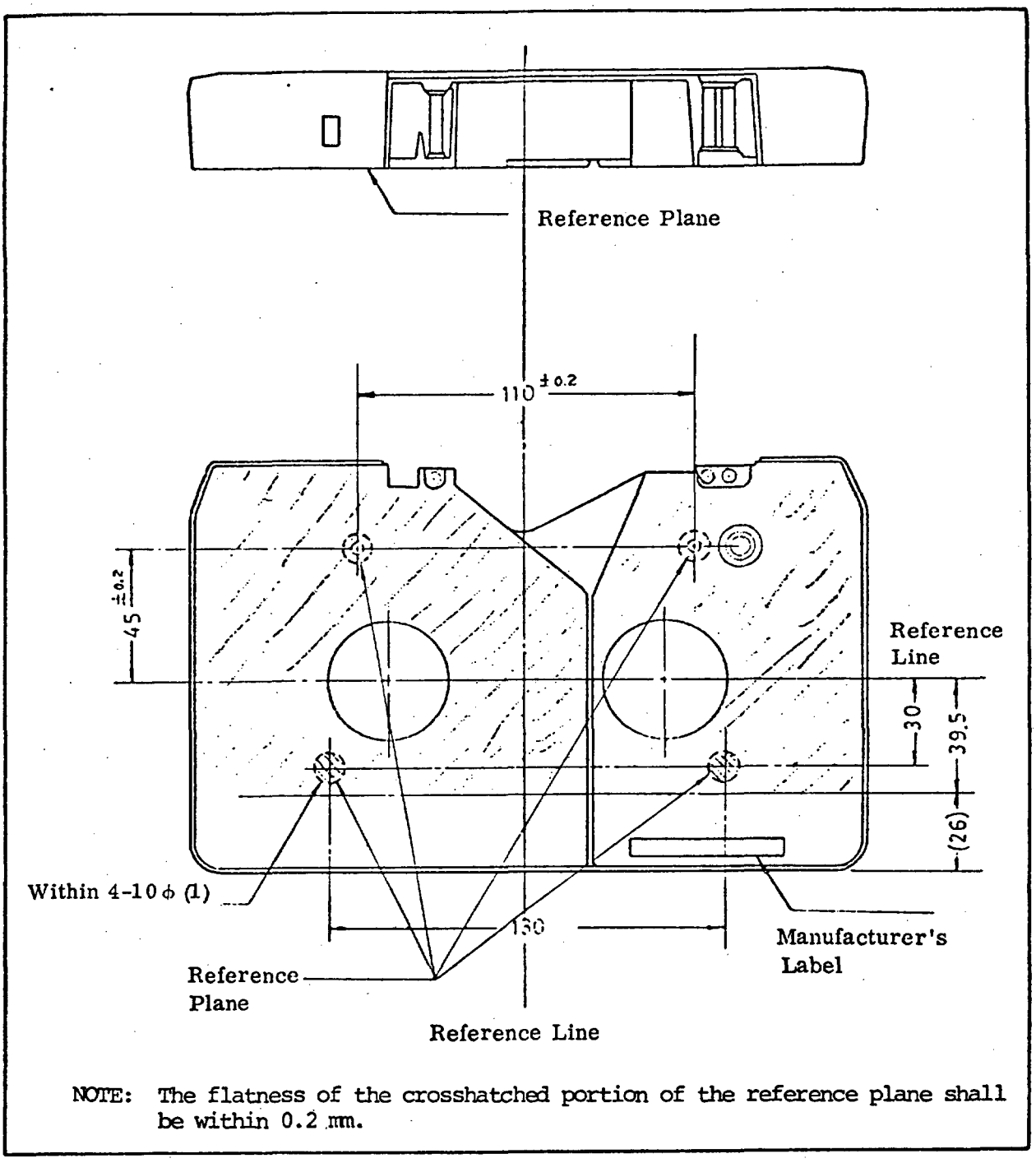


Figure 6. THREE-FOURTHS-INCH MAGNETIC TAPE CASSETTE.

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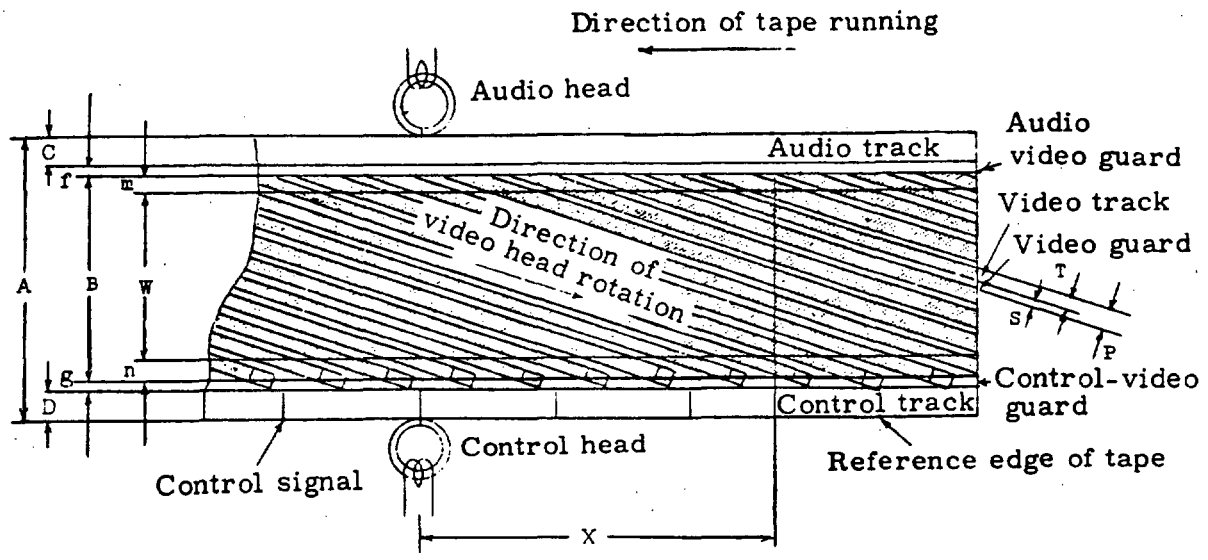


NOTE: The flatness of the crosshatched portion of the reference plane shall be within 0.2 mm.

Figure 7. THREE-FOURTHS-INCH MAGNETIC TAPE CASSETTE.

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MAGNETIC TAPE FORMAT



NOTE: The figure shows the magnetic coated surface of the tape.

Symbol	(in mm)
A	$12.7^{+0.0}_{-0.1}$
B	10.65
C	$1.0^{+0.0}_{-0.1}$
f	0.15
m	0.275
w	10.10
n	0.275
g	0.1
D	0.8
P	0.173
S	more than 0.04
T	more than 0.1
X	81.0

REMARKS:

(1) The video track can be recorded in the control-video guard band.

(2) It is allowable if a part of the "scan beginning overlap width (m)" or a part of the "scan ending overlap width (n)" is missing from the video track. But the range of the video width (W) shall be correct.

Figure 8. ONE-HALF-INCH, HELICAL SCAN, REEL-TO-REEL, FORMAT FOR.

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TABLE II

Symbol	Item		
A	TAPE WIDTH	mm	$12.7\ ^{+0}_{-0.1}$
Vt	TAPE SPEED	mm/sec	190.5 ±0.5%
ϕ	DRUM DIAMETER	mm	115.82 ±0.01
Vh	WRITING SPEED	m/sec	11.1
P	VIDEO TRACK PITCH	mm	0.173
B	FULL VIDEO WIDTH	mm	10.65
W	VIDEO WIDTH	mm	10.10
C	AUDIO TRACK WIDTH	mm	1.0
D	CONTROL TRACK WIDTH	mm	0.8
f	AUDIO-VIDEO GUARD WIDTH	mm	0.15
g	CONTROL-VIDEO GUARD WIDTH	mm	0.1
m	SCAN BEGINNING OVERLAP WIDTH	mm	0.275
n	SCAN ENDING OVERLAP WIDTH	mm	0.275
θ_0	VIDEO TRACK ANGLE (STOPPING)		$3^{\circ}11'$
θ	VIDEO TRACK ANGLE (RUNNING)		$3^{\circ}7'43"$
ℓ	VIDEO TRACK LENGTH	mm	185.1
X	POSITIONS OF AUDIO AND CONTROL HEADS	mm	81.0

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